

Estimating parameters

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Goal

Logistic regression model:

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 X_i$$

Given data $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$, how do I estimate β_0 and β_1 ?

Motivating example

Y_i = result of flipping a coin (Heads or Tails)

Is Y_i a random variable?

Motivating example

Y_i = result of flipping a coin (Heads or Tails)

Let's make a model:

+ **Step 1:** Distribution of the response

$$Y_i \sim \text{Bernoulli}(\pi)$$

+ **Step 2:** Construct a model for the parameters

$$\pi = ??$$

Motivating example

Y_i = result of flipping a coin (Heads or Tails)

Suppose your friend estimates that the probability of heads is 0.9

+ $Y_i \sim \text{Bernoulli}(\pi)$

+ $\hat{\pi} = 0.9$

How can we assess whether this estimate $\hat{\pi}$ is reasonable?

Motivating example

Suppose we flip the coin 5 times, and observe

$$y_1, \dots, y_5 = T, T, T, T, H$$

What is the probability of (i.e., how *likely* is) getting this string of flips if $\pi = 0.9$? Discuss with your neighbor for a minute, then we will discuss as a class.

Likelihood

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

- + Model: $Y_i \sim Bernoulli(\pi)$, and $\hat{\pi} = 0.9$
- + Data: $y_1, \dots, y_5 = T, T, T, T, H$
- + Likelihood: $L(\hat{\pi}) = P(y_1, \dots, y_5 | \pi = 0.9) = 0.00009$

Class Activity, Part I

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class Activity

$$\begin{aligned} L(0.2) &= P(y_1, \dots, y_5 | \pi = 0.2) \\ &= (0.2)(0.8)(0.8)(0.2)(0.8) = 0.020 \end{aligned}$$

$$\begin{aligned} L(0.3) &= P(y_1, \dots, y_5 | \pi = 0.3) \\ &= (0.3)(0.7)(0.7)(0.3)(0.7) = 0.031 \end{aligned}$$

Which value, 0.2 or 0.3, seems more reasonable?

Class Activity

Which value of $\hat{\pi}$ in the table would you pick?

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

$\hat{\pi}$	Likelihood
0.30	0.031
0.35	0.033
0.40	0.036
0.45	0.033

Maximum likelihood estimate: $\hat{\pi} = 0.4$

Maximum likelihood

Maximum likelihood principle: estimate the parameters to be the values that maximize the likelihood

Steps for maximum likelihood estimation:

- + *Likelihood*: For each potential value of the parameter, compute the likelihood of the observed data
- + *Maximize*: Find the parameter value that gives the largest likelihood

Maximum likelihood for logistic regression

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Size}_i \quad \pi_i = \frac{\exp\{\beta_0 + \beta_1 \text{Size}_i\}}{1 + \exp\{\beta_0 + \beta_1 \text{Size}_i\}}$$

Observed data:

Tumor cancerous	Yes	No	No	Yes	No
Size of tumor (cm)	6	1	0.5	4	1.2

Suppose $\beta_0 = -2$, $\beta_1 = 0.5$. How would I compute the likelihood?

Class Activity, Part II

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class Activity

$$\hat{\pi}_i = \frac{\exp\{-2 + 0.5 \text{ Size}_i\}}{1 + \exp\{-2 + 0.5 \text{ Size}_i\}}$$

Tumor cancerous	Yes	No	No	Yes	No
Size of tumor (cm)	6	1	0.5	4	1.2
$\hat{\pi}_i$					

Likelihood =

Maximum likelihood for logistic regression

Likelihood:

- + For estimates $\hat{\beta}_0$ and $\hat{\beta}_1$, $\hat{\pi}_i = \frac{\exp\{\hat{\beta}_0 + \hat{\beta}_1 X_i\}}{1 + \exp\{\hat{\beta}_0 + \hat{\beta}_1 X_i\}}$
- + $L(\hat{\beta}_0, \hat{\beta}_1) = P(Y_1, \dots, Y_n | \hat{\beta}_0, \hat{\beta}_1)$

Maximize:

- + Choose $\hat{\beta}_0, \hat{\beta}_1$ to maximize $L(\hat{\beta}_0, \hat{\beta}_1)$

So far, we only considered a few values for β_0 and β_1 . How should we check other values, to make sure our estimates actually maximize likelihood?

Computing likelihood in R

Observed data: T, T, T, T, H

- + We are going to consider several different potential values for $\hat{\pi}$:

$$0, 0.1, 0.2, 0.3, \dots, 0.9, 1$$

- + For each potential value, we will compute the likelihood:

$$L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$$

- + We then see which value has the highest likelihood.
- + Is this all possible values? No, but let's start here.

R code

```
# List the values for pi hat  
pi_hat <- seq(from = 0, to = 1, by = 0.1)
```

R code

```
# List the values for pi hat  
pi_hat <- seq(from = 0, to = 1, by = 0.1)  
  
# Create a space to store the likelihoods  
likelihood <- rep(0, length(pi_hat))
```

R code

```
# List the values for pi hat
pi_hat <- seq(from = 0, to = 1, by = 0.1)

# Create a space to store the likelihoods
likelihood <- rep(0,length(pi_hat))

# Compute and store the likelihoods
for( i in 1:length(pi_hat) ){
  likelihood[i] <- pi_hat[i]*(1-pi_hat[i])^4
}
```

Results

pi_hat	likelihood
0.0	0.00000
0.1	0.06561
0.2	0.08192
0.3	0.07203
0.4	0.05184
0.5	0.03125
0.6	0.01536
0.7	0.00567
0.8	0.00128
0.9	0.00009
1.0	0.00000